

▼ **Figure 34.36 Feet adapted to perching.** This great tit (*Parus major*) is a member of the Passeriformes, the perching birds. The toes of these birds can lock around a branch or wire, enabling the bird to rest for long periods.



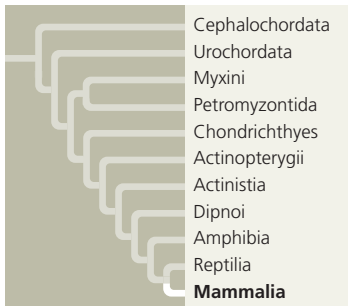
CONCEPT CHECK 34.5

1. Describe three key amniote adaptations for life on land.
2. Are snakes tetrapods? Explain.
3. Identify four avian adaptations for flight.
4. **VISUAL SKILLS** Based on the phylogeny shown in Figure 34.25, identify the sister group for (a) reptiles, (b) squamates, and (c) the clade that includes crocodilians and birds.

For suggested answers, see Appendix A.

CONCEPT 34.6

Mammals are amniotes that have hair and produce milk

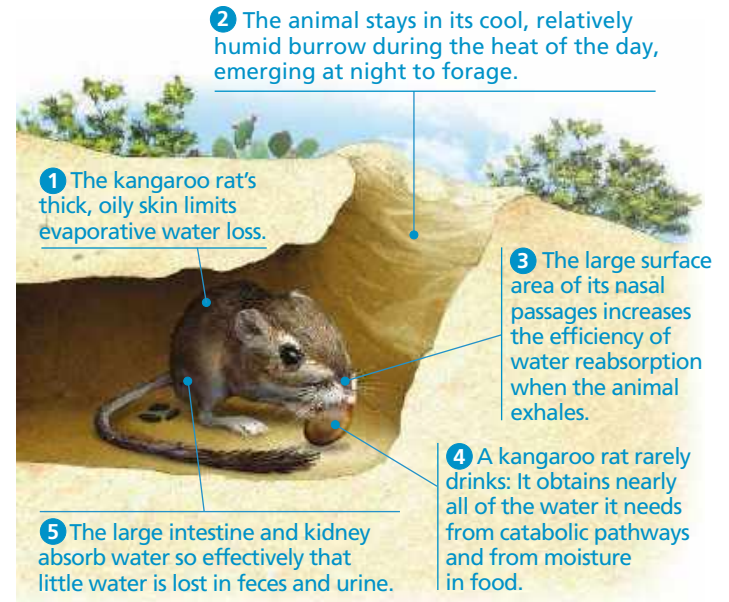


The reptiles we have been discussing represent one of the two living lineages of amniotes. The other amniote lineage is our own, the **mammals**. Today, there are about 6,400 known species of mammals on Earth.

Derived Characters of Mammals

Mammals are named for their distinctive mammary glands, which produce milk for offspring. All mammalian mothers nourish their young with milk, a balanced diet rich in fats, sugars, proteins, minerals, and vitamins. Hair, another mammalian character, and a fat layer under the skin provide insulation that can conserve water and protect the body against extremes of heat or cold. Another mammalian adaptation for life on land is the kidney (see Figure 44.13), which is efficient at conserving water when removing wastes from the body. Some mammals, such as kangaroo rats, are so adept at conserving water that they can survive

▼ **Figure 34.37 Adaptations of the kangaroo rat to its extremely dry habitat.**



MAKE CONNECTIONS Explain how the catabolic pathways mentioned in 4 could provide a kangaroo rat with water. (See Concept 9.1.)

in arid environments while drinking little or no water at all (**Figure 34.37**).

Like birds, mammals are endothermic, and most have a high metabolic rate. Efficient respiratory and circulatory systems (including a four-chambered heart) support a mammal's metabolism. Also as in birds, mammals generally have a larger brain than other vertebrates of equivalent size, and many species are capable learners. The relatively long duration of parental care extends the time for offspring to learn important survival skills by observing their parents. In addition, whereas the teeth of reptiles are generally uniform in size and shape, the jaws of mammals typically bear a variety of teeth with sizes and shapes adapted for chewing many kinds of foods. Humans, like most mammals, have teeth modified for shearing (incisors and canine teeth) and for crushing and grinding (premolars and molars).

Early Evolution of Mammals

Mammals belong to a group of amniotes known as **synapsids**. Early nonmammalian synapsids lacked hair, had a sprawling gait, and laid eggs. A distinctive characteristic of synapsids is the single temporal fenestra, a hole behind the eye socket on each side of the skull. Humans retain this feature; your jaw muscles pass through the temporal fenestra and anchor on your temple. Fossil evidence shows that the